

```
// Go through the resulting list
for each power_source in power_source_instances:
    If WIRED feature is present and power_source.Status == Active:
        return false
    If BAT feature is present and power_source.Status == Active:
        return true

// No definitive information was found so assume it's not battery powered
return false
```

11.4.4. SFrame End-to-End Encryption

To allow for cameras to be deployed in complex ecosystems where high bitrate media streams only need to be transmitted out of the home once, the [Secure Frame \(SFrame\)](#) protocol SHALL be supported by a device.

SFrame provides an end-to-end encryption and authentication mechanism for media frames in a multi-party conference call, in which central media servers (Selective Forwarding Units or SFUs) can access the media metadata needed to make forwarding decisions, without having access to the actual media.

While SFrames are not yet part of the core WebRTC standard when this cluster's initial release occurred, the W3C has designed a mechanism that allows them to be supported within web browsers and other WebRTC compliant devices via [W3C Encoded Transforms](https://www.w3.org/TR/webrtc-encoded-transform/) [https://www.w3.org/TR/webrtc-encoded-transform/]. This specification will state how cameras are to implement SFrame and its key distribution so that any device or ecosystem can implement end-to-end encryption.

The IETF has begun the standardization process for SFrame support inside RTP and SDP in conjunction with the W3C work. Servers SHALL support SFrame in RTP via the process identified in [draft-ietf-avtcore-rtp-sframe](https://datatracker.ietf.org/doc/html/draft-ietf-avtcore-rtp-sframe) [https://datatracker.ietf.org/doc/html/draft-ietf-avtcore-rtp-sframe].

Use of SFrame for any WebRTC session is optional.

Within the overall design of SFrame, the Matter CASE session provides the preexisting secure signaling channel to exchange the values for the SFrame KID, base_key, and cipher suite.

11.4.4.1. KID and Ratcheting

Devices SHALL use a ratchet step size R of 8 bits. This leaves 8 to 56 bits for use as the "key generation" value. The resulting KID will always be at least 2 bytes long.

Providers SHOULD ratchet their keys at least once every 10 minutes using the algorithm described in [Section 5.1](https://www.rfc-editor.org/rfc/rfc9605.html#name-sender-keys) [https://www.rfc-editor.org/rfc/rfc9605.html#name-sender-keys] of [RFC 9605](#). Devices SHALL keep the previous ratchet key iteration around for one ratchet period to handle any in-flight or retransmissions on the two-way talk channel.

Requestors SHALL ratchet their sending keys when they detect a ratchet by the Provider.